Mitigating Ill-Posedness in SAR Ionospheric Tomography Toward Multisatellite Collaboration

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***Abstract*—In recent years, spaceborne low-frequency fully polarimetric synthetic aperture radar (PolSAR) ionospheric tomography has been an emerging technology for 3-D ionospheric computerized tomography (CIT). However, conventional single-satellite SAR (e.g., PALSAR) tomography suffers from pronounced ill-posedness due to restricted observation geometry, resulting in low reconstruction accuracy. Motivated by this issue and the future trend of multisatellite collaborative observation, where each satellite operates in a similar low-frequency fully polarimetric mode (e.g., PALSAR-2/3, BIOMASS SAR, and NISAR), this article proposes a novel joint 3-D reconstruction framework for low-frequency PolSAR CIT. A semi-physical simulation framework is established based on historical PALSAR-1 data to construct a robust ray observation geometry. Mathematical evaluations demonstrate that the multisatellite configuration significantly reduces the condition number of the geometric matrix, mitigating the ill-posedness of the tomographic equations. Furthermore, by constructing an augmented state vector, the proposed method adaptively isolates and calibrates heterogeneous platform-specific polarimetric system errors, enabling reliable absolute total electron content (TEC) estimation without relying on external Global Navigation Satellite System (GNSS) references. The 3-D reconstruction results reveal that, compared to the conventional single-satellite model, the root-mean-square error (RMSE) is reduced by 12.2%, and the condition number of the normal matrix (associated with the tomographic inversion) is reduced by a factor of 1.63, demonstrating a measurable enhancement in the numerical stability of the multisatellite configuration. This study can provide vital theoretical support and an algorithmic reference for 3-D ionospheric sensing in future L-band multisatellite SAR constellations.**

***Index Terms*—Computerized ionospheric tomography, electron density, ill-posedness, PALSAR, polarimetric synthetic aperture radar (PolSAR).**

I. Introduction

THE ionosphere significantly impacts the propagation of spaceborne microwave signals, introducing propagation delays and phase shifts that are critical for navigation and remote sensing [1], [2]. In recent years, spaceborne fully polarimetric synthetic aperture radar (PolSAR) has emerged as a powerful tool for ionospheric remote sensing [3], [4]. By measuring the Faraday rotation angle (FRA) derived from the polarimetric scattering matrix, PolSAR enables the extraction of total electron content (TEC) with unprecedented high spatial resolution [5], [6]. This capability is particularly useful for high-latitude regions (e.g., glaciers and permafrost zones). In these remote areas, the sparse or entirely absent deployment of ground-based Global Navigation Satellite System (GNSS) receivers renders traditional GNSS ionospheric computerized tomography (CIT) unfeasible [7], [8]. Consequently, spaceborne PolSAR CIT stands as a crucial technique to fill the ionospheric information gap in these harsh environments.

Extensive research has established that the Bickel–Bates (B&B) algorithm [9] is widely regarded as the gold standard for FRA retrieval, owing to its theoretical immunity to polarimetric amplitude imbalance. On this basis, Meyer and Nicoll [10] implemented the first complete FRA prediction, detection, and correction workflow based on the B&B framework for spaceborne L-band polarimetric SAR data, verifying its engineering practicability. Rogers and Quegan [11] quantitatively confirmed the high accuracy and robustness of the B&B algorithm under nonideal system conditions through systematic multimethod evaluation. Quegan and Lomas [12] further improved the theoretical error system of the B&B algorithm by deriving the explicit expression of FRA estimation bias caused by system imperfections. Kim et al. [13] extended the core framework of the B&B algorithm to the correction of polarimetric SAR distortion induced by ionospheric scintillation, greatly broadening its application boundary. However, applying this algorithm to high-latitude ionospheric sensing presents a severe physical challenge. In high-latitude regions, the near-parallel alignment of the radar line-of-sight and geomagnetic field lines generates a significantly large FRA. Theoretical error modeling reveals that the B&B estimator remains highly susceptible to polarimetric crosstalk and resid-

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ual phase imbalance, and this systematic bias is amplified by the sin(2 Ω) term in high-latitude environments. This introduces a nonnegligible, heterogeneous systematic bias into the TEC observations. Consequently, traditional CIT relying on a single satellite platform becomes severely ill-posed [14], [15], suffering from rank deficiency, which fails to decouple the electron density from hardware-induced errors.

To mitigate the ill-posedness of single-pass tomography, multisatellite collaborative observation provides crucial geometric redundancy. Nevertheless, directly fusing observations from heterogeneous platforms (e.g., PALSAR-1 [10] and PALSAR-2 [16]) is problematic. Since different platforms exhibit distinct hardware isolation levels, their residual systematic biases are heterogeneous. Forcing a conventional linear inversion on such conflicting TEC measurements inevitably yields severe reconstruction artifacts and geometric distortions, obliterating the fine-scale ionospheric structures over polar regions.

To address these critical challenges, this study proposes a joint tomographic reconstruction framework for CIT using multisatellite PolSAR. By modeling the heterogeneous systematic biases as extended state variables, we construct an augmented tomographic matrix that leverages the intersecting geometry of radar rays within the voxels. The inversion is constrained by priors from the Empirical Canadian High Arctic Ionospheric Model (E-CHAIM) [17], enabling the framework to adaptively isolate and calibrate the platform-specific biases.

The primary contribution of this work lies in the establishment of a unified analytical error model. This model rigorously quantifies the mapping mechanisms from polarimetric crosstalk, phase imbalance, and thermal noise to absolute TEC observation errors. In particular, it reveals a physical amplification effect of systematic biases in high-latitude environments, where the radar line-of-sight aligns nearly parallel to the geomagnetic field lines. Building upon this theoretical foundation, we transform the intractable heterogeneous bias problem into a mathematically solvable joint estimation task, enabling self-adaptive hardware calibration without external referencing.

Although several L-band PolSAR satellites (e.g., PALSAR-2/3, SAOCOM-1A/B [18], and LT-1A/B [19]) are currently in orbit, these missions are primarily designed for Earth surface observation rather than dedicated 3-D ionospheric sounding. Consequently, their independent orbital configurations and task scheduling make it practically infeasible to obtain strictly synchronized, simultaneous multiangle overlapping acquisitions over the same region, which is a strict spatio-temporal prerequisite for instantaneous ionospheric tomography. In practical multisatellite SAR CIT, temporal consistency is as important as spatial overlap. Strictly, the electron density field should remain nearly unchanged during the acquisition interval of different satellite passes; otherwise, ionospheric temporal evolution will enter the TEC observation equation as an additional model error. Therefore, the semi-physical simulation in this study is interpreted as a quasi-synchronous experiment under a frozen-ionosphere assumption at a reference epoch. This assumption allows us to focus on the geometric conditioning and platform-dependent bias calibration of the proposed framework. For real-time asynchronous acquisitions, the data should be selected within a short temporal window under relatively quiet ionospheric conditions. Small residual temporal mismatches can be treated as additional observation uncertainty and incorporated into the weighting matrix, whereas long time gaps or disturbed ionospheric conditions would require a dynamic reconstruction scheme with temporal regularization. Therefore, to overcome these physical observational constraints and provide a robust algorithmic validation for future collaborative SAR constellations, a reliable semi-physical simulation framework is established. Comprehensive numerical simulations are conducted using full-pol PALSAR-1 historical data [20] parameters to demonstrate the superiority and feasibility of the proposed joint reconstruction method.

The outline of this article is given as follows. In Section II, the theoretical modeling and error analysis, and the multisatellite joint CIT framework are introduced. In Section III, comprehensive numerical simulations and performance evaluations are conducted using full-pol PALSAR data parameters to demonstrate the superiority and feasibility of the proposed algorithm. Finally, our conclusion is drawn in Section IV.

II. Methodology

A. Theoretical Modeling and Error Analysis

To achieve reliable CIT using spaceborne PolSAR, it is imperative to establish a rigorous mapping mechanism from raw polarimetric measurements to the final TEC observations. This section formulates a unified analytical error model, strictly decoupling the impacts of the ideal FRA, deterministic hardware biases, and stochastic thermal noise.

1) Theoretical Analysis of the B&B Algorithm: To derive the ideal B&B observation relationship, the scattering medium is first assumed to satisfy the reciprocity theorem ($S_{HV} = S_{VH}$), and radar system errors and thermal noise are temporarily omitted from the ideal model. These error sources are then explicitly reintroduced in the subsequent bias and noise models. The measured polarimetric scattering $\mathbf{M}$ in the linear horizontal and vertical (H, V) polarization basis, which is subjected to two-way ionospheric propagation characterized by the one-way FRA, denoted by Ω, can be modeled as

$$\begin{bmatrix} M_{HH} & M_{VH} \\ M_{HV} & M_{VV} \end{bmatrix} = \begin{bmatrix} \cos\Omega & \sin\Omega \\ -\sin\Omega & \cos\Omega \end{bmatrix} \cdot \begin{bmatrix} S_{HH} & S_{VH} \\ S_{HV} & S_{VV} \end{bmatrix} \cdot \begin{bmatrix} \cos\Omega & \sin\Omega \\ -\sin\Omega & \cos\Omega \end{bmatrix} \tag{1}$$

where $S_{p,q}$ $(p, q \in \{H, V\})$ denotes the true complex scattering amplitude of the target.

The B&B algorithm projects this linear scattering matrix into a circular polarization basis (L, R) via the unitary transformation matrix [9]

$$T = 1 \Big/ \sqrt{2} \begin{bmatrix} 1 & 1 \\ \mathrm{j} & -\mathrm{j} \end{bmatrix}. \tag{2}$$

The circular scattering matrix $\mathbf{Z}$ is obtained by $\mathbf{Z} = \mathbf{T}^{-1}\mathbf{M}\mathbf{T}$. In the ideal noise-free and perfectly calibrated scenario, the circular cross-polarization terms rigorously encode the FRA

$$Z_{LR} = Z_X e^{\mathrm{j}2\Omega}, \quad Z_{RL} = Z_X e^{\mathrm{j}2\Omega} \tag{3}$$

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TABLE I
HARDWARE BIAS AND NOISE INJECTION PARAMETERS FOR THE HETEROGENEOUS SAR CONSTELLATION

Platform	Benchmarked System	δ	ϕ	SNR
Sat A	PALSAR-2	-30 dB	3.0°	20 dB
Sat B	PALSAR-3	-35 dB	2.0°	22 dB
Sat C	SAOCOM	-30 dB	2.0°	18 dB

where $Z_x = \mathrm{j}/2(S_{HH} + S_{VV})$. Thus, the B&B estimator extracts Ω through the complex cross-product $A = Z_{LR}Z_{RL}^*$

$$\hat{\Omega}_{ideal} = \frac{1}{4}\arg\left(Z_{LR}Z_{RL}^*\right). \tag{4}$$

2) Analytical Derivation of Heterogeneous Systematic Biases: In practical multisatellite operations, the PolSAR system inevitably suffers from imperfect hardware isolation. We employ first-order complex perturbation theory to derive the analytical bias introduced by polarimetric crosstalk and channel imbalance.

The assumptions used in the following derivation are made to isolate the dominant first-order platform-dependent error terms. Their validity is supported by the hardware-error settings listed in Table I. The crosstalk levels of −30 to −35 dB correspond to $|\delta| = 0.0178$–0.0316, while the residual phase imbalance of 2°–3° corresponds to $|\phi| = 0.0349$–0.0524 rad. Therefore, the neglected second-order terms are much smaller than the retained first-order terms, with $|\delta|^2 \le 1.0 \times 10^{-3}$, $|\phi|^2 \le 2.7 \times 10^{-3}$, and $|\delta\phi| \le 1.7 \times 10^{-3}$. These quantities are sufficiently small for a first-order perturbation analysis and do not affect the leading bias mechanism discussed here. The thermal noise is not neglected in the final observation model; it is treated separately through the Cramer–Rao lower bound (CRLB)-based variance model and the weighting matrix W in Sections II-A3 and II-B.

Crosstalk: Let δ represent the symmetric complex crosstalk coefficient ($\delta \ll 1$). The perturbed circular cross-polarization measurements, omitting second-order negligible terms, are given by $\hat{Z}_{LR} = Z_{LR} + \Delta Z_{LR}$, where the perturbation term is dominated by the leakage from the strong co-polarization signals

$$\Delta Z_{LR} = \Delta Z_{RL} \approx -\mathrm{j}\delta\left(S_{HH} - S_{VV}\right). \tag{5}$$

Defining the co-polarization ratio of the target as

$$\alpha = \frac{S_{HH} - S_{VV}}{S_{HH} + S_{VV}} \tag{6}$$

the complex perturbation ratio of the B&B core product A can be derived as

$$\varepsilon/A \approx -4Re\left(\delta\alpha\right)e^{-\mathrm{j}2\Omega}. \tag{7}$$

Applying the small-angle phase approximation

$$\arg\left(A + \varepsilon\right) \approx \arg\left(A\right) + Im\left(\varepsilon/A\right). \tag{8}$$

The systematic bias induced by crosstalk is

$$\Delta\Omega_{cross} \approx \mathrm{Re}\left(\delta\alpha\right)\sin\left(2\ \Omega\right). \tag{9}$$

The derivation of (9) is provided in Appendix A.

Channel Imbalance: Let Δ_A denote the relative imbalance of the V channel with respect to the H channel. The induced perturbation in the circular basis is approximately

$$\Delta Z_{LR} = \Delta Z_{RL} \approx -\frac{1}{2}\Delta_A S_{VV}. \tag{10}$$

Setting the target parameter $\beta = S_{VV}/S_{HH} + S_{VV}$, the phase error term relies on the variable $Q = j\Delta_A B$. Crucially, this derivation mathematically proves the unique properties of the B&B algorithm: If the imbalance is purely a polarimetric amplitude imbalance (Δ_A is real), Q becomes a pure imaginary number, resulting in its real part being zero. Consequently, the phase error vanishes entirely, proving the B&B algorithm's absolute theoretical immunity to amplitude imbalance.

Conversely, if the system exhibits a residual phase imbalance ($\Delta_A = j\phi$, where ϕ is real), Q becomes a real number ($-\phi\beta$). This introduces a severe systematic bias strictly analogous to the crosstalk effect

$$\Delta\Omega_{imb} \approx \frac{1}{2}\phi\beta\sin\left(2\ \Omega\right). \tag{11}$$

Therefore, the combined compound systematic bias for the ith radar ray observed by a specific satellite platform is defined as

$$B_{\Omega,i} \approx \left[Re\left(\delta_i\alpha\right) + \frac{1}{2}\phi_i\beta\right]\sin\left(2\ \Omega\right) \tag{12}$$

where i (i = 1, 2,…, I) denotes the index of the radar ray, and I is the total number of valid observations collectively acquired by the multisatellite network. The derivation of (12) is provided in Appendix B. It must be emphasized that because different satellite platforms possess completely distinct hardware aging and thermal deformations, their δ_i and ϕ_i are fundamentally different, rendering the overall system bias $B_{\Omega,i}$ highly heterogeneous across the multisatellite network.

3) Thermal Noise and Statistical Variance: Beyond deterministic biases, the PolSAR system is subject to additive thermal white noise $\mathbf{N}$, which induces zero-mean random phase variance. By analogizing the complex product of circular cross-polarization terms to a zero-baseline interferometric pair, we characterize the FRA estimation precision using multilook interferometric statistics. Within this statistical framework, the phase variance is primarily governed by the degree of coherence between the Z_{LR} and Z_{RL} Channels, which is intrinsically tied to the system's thermal signal-to-noise ratio (SNR). The SNR can be defined as

$$SNR = \frac{E\left[|Z_x|^2\right]}{\sigma_n^2} \tag{13}$$

where E[·] denotes the mathematical expectation, which represents the statistical average over a finite number of looks in multilook processing. σ_n^2 can be characterized by the noise-equivalent sigma zero (NESZ) of the SAR system.

Specifically, in multilook processing with an equivalent number of looks (ENLs), L, the phase variance σ_Ω^2 is strictly constrained by the CRLB

$$\sigma_\Omega^2 = \frac{1}{32L}\frac{1-\gamma^2}{\gamma^2} \tag{14}$$

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where the complex coherence coefficient γ represents the cross-channel correlation. In the absence of polarimetric crosstalk, γ is fundamentally determined by the SNR as

$$\gamma \approx \frac{\mathrm{SNR}}{1+\mathrm{SNR}}. \tag{15}$$

This statistical property indicates a critical divergence in estimation uncertainty as the SNR diminishes. In high-latitude study areas, the 3-D tomographic system faces a dual challenge: the deterministic systematic biases are significantly amplified by the large sin (2 Ω) term, while the stochastic errors are further exacerbated by the potential decrease of coherence γ. Specifically, in regions characterized by weak surface backscattering, such as smooth glacial surfaces or certain permafrost zones, the diminished SNR leads to a rapid increase in the phase variance σ_{Ω}^2. This phenomenon intensifies the ill-posedness of the single-satellite inversion, making it increasingly difficult to achieve a stable reconstruction of the ionospheric electron density without additional geometric constraints.

4) Unified Observation Equation: Finally, mapping the FRA from the phase domain to the physical TEC domain requires the standard magneto-ionic relationship [21]

$$\mathrm{TEC} = \frac{\omega_0^2 \Omega_{B\&B}}{9.327 \cdot 10^5 B_0 \cos\theta_B} \tag{16}$$

where ω_0 represents the carrier frequency expressed in radians, B_0 is the strength of the ambient geomagnetic field, and θ_B denotes the angle between the radio wave vector and the geomagnetic field vector.

Synthesizing the ideal projection, the heterogeneous systematic bias, and the random thermal noise, the unified absolute TEC observation equation for the ith single ray can be rigorously formulated as

$$Y_i = \mathrm{TEC}_{true,i} + B_{\Omega,i} + E_i\left(0, \sigma_{TEC}^2\right) \tag{17}$$

where Y_i represents the absolute TEC value extracted via the PolSAR-B&B estimator. B_i denotes the heterogeneous systematic bias, and E_i is the zero-mean random noise term whose variance is derived from the statistical model. This additive model serves as the foundational equation for the subsequent multisatellite joint tomographic inversion.

B. Multisatellite Joint Tomographic Framework

Based on the unified observation model derived in Section II-A, traditional single-platform CIT suffers from severe rank deficiency and is highly sensitive to platform-specific hardware biases. To achieve more reliable 3-D electron density mapping in high-latitude regions, we propose an augmented state joint inversion framework that algebraically decouples the ionospheric structures from heterogeneous hardware biases.

Consider a multisatellite collaborative network comprising K heterogeneous PolSAR platforms (e.g., $K = 3$ in this study), which collectively acquire I valid radar rays penetrating a predefined 3-D tomographic grid with N voxels. The traditional state vector $\mathbf{x} = [x_1, x_2, \ldots, x_N]^T \in \Re^N$ represents the unknown electron density (Ne) in each voxel. To adaptively calibrate the hardware biases without external GNSS references, we define a bias vector $\mathbf{b} = [b_1, b_2, \ldots, b_K]^T \in \Re^K$ and construct an augmented state vector $x_{aug} \in \Re^{N+K}$, i.e.,

$$x_{aug} = \begin{bmatrix} \mathbf{x} \\ \mathbf{b} \end{bmatrix}. \tag{18}$$

Correspondingly, the unified observation system can be expressed in a compact algebraic matrix form

$$\mathbf{Y} = [\mathbf{G} \quad \mathbf{H}] \begin{bmatrix} \mathbf{x} \\ \mathbf{b} \end{bmatrix} + \mathbf{e} = \mathbf{G}_{aug}\mathbf{x}_{aug} + \mathbf{e} \tag{19}$$

where $\mathbf{Y} \in R^{I\times 1}$ is the absolute TEC observation vector for all valid rays, $\mathbf{x} \in R^{N\times 1}$ is the unknown electron density vector of the N voxels, and $\mathbf{b} \in R^{K\times 1}$ is the platform-dependent bias vector for the K satellite platforms. $\mathbf{G} \in R^{I\times N}$ is the geometric design matrix whose elements describe the ray-voxel intersection lengths, and $\mathbf{H} \in R^{I\times K}$ is a sparse platform-indicator matrix. Specifically, $H_{ik} = 1$ if the ith ray is acquired by the kth satellite platform, and $H_{ik} = 0$ otherwise. Therefore, the augmented matrix is $\mathbf{G}_{aug} = [\mathbf{G} \quad \mathbf{H}] \in R^{\{I\times(N+K)\}}$, and the augmented state vector is $\mathbf{x}_{aug} = [\mathbf{x}^T \ \mathbf{b}^T]^T \in R^{\{(N+K)\times 1\}}$. The noise vector $\mathbf{e} \in R^{(I\times 1)}$ represents the random observation error.

To account for the severe spatial variance of the stochastic noise $E_i \sim N(0, \sigma_{TEC,i}^2)$ under low SNR conditions, we introduce a diagonal statistical weight matrix $\mathbf{W} \in \Re^{I\times I}$. Its diagonal elements are strictly defined by the inverse of the analytical variance derived from the multilook interferometric statistics

$$\mathbf{W}_{i,i} = 1/\sigma_{TEC,i}^2. \tag{20}$$

For time-asynchronous observations, the variance term in W can be generalized to include both thermal noise and temporal mismatch errors caused by ionospheric evolution between the acquisition time and the reference reconstruction epoch. This weighting strategy dynamically suppresses the contribution of severely noise-corrupted rays (e.g., those reflected from smooth glacial surfaces) while amplifying the high-coherence observations.

Mathematically, the ill-posedness of the tomographic observation geometry can be rigorously quantified by the condition number κ of the design matrix. Based on the singular value decomposition (SVD), the condition number is defined as the ratio of the maximum singular value ($\sigma_{\max}$) to the minimum singular value (σ_{mi}), i.e., $\kappa(G) = \sigma_{\max}/\sigma_{\min}$. A severely ill-posed single-satellite geometric matrix $\mathbf{G}$ typically exhibits a rapid singular value spectrum decay and an extremely large κ, implying that the inversion is highly vulnerable to measurement noise and system biases [22], [23]. By constructing the augmented matrix G_{aug} via multisatellite collaborative observations, the spatial intersection redundancy can be enhanced, which is theoretically expected to flatten the singular value spectrum and effectively reduce the condition number.

Furthermore, given the inherent ill-posedness of the intersecting geometry at high latitudes, Tikhonov regularization is introduced using the E-CHAIM as the background a priori constraint. Ultimately, the multisatellite joint tomographic

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problem is mathematically transformed into minimizing the following weighted regularized objective function:

$$J\left(\mathbf{x}_{aug}\right) = \left(\mathbf{Y} - \mathbf{G}_{aug}\mathbf{x}_{aug}\right)^{T} \mathbf{W}\left(\mathbf{Y} - \mathbf{G}_{aug}\mathbf{x}_{aug}\right) + \lambda \|\mathbf{D}\left(\mathbf{x} - \mathbf{x}_{prior}\right)\|_2^2 \quad (21)$$

where $\mathbf{x}_{prior} \in \Re^N$ is the background electron density vector provided by E-CHAIM, $\mathbf{D} \in \Re^{N\times N}$ is the spatial smoothing operator (e.g., a discrete Laplacian matrix), and λ is the regularization parameter balancing the data fidelity and the a priori physical constraints. In this study, λ was determined after normalizing the augmented observation equations. A group of logarithmically spaced candidate values was tested, and the tradeoff between the observation residual and the E-CHAIM-constrained regularization term was examined following the L-curve criterion. The value near the turning region of this tradeoff curve was selected as $\lambda 0$. This setting avoids an excessively weak prior constraint, which may cause leakage between ionospheric perturbations and platform-dependent biases, while also preventing the solution from being dominated by the background model. The same λ-selection strategy was used throughout the comparative experiments. By minimizing $J(\mathbf{x}_{aug})$, the proposed framework simultaneously yields the true 3-D ionospheric structures and calibrates the heterogeneous systematic biases. The Tikhonov-regularized augmented inversion is used to estimate the platform-dependent bias terms and to stabilize the observation model with the E-CHAIM prior. After this calibration step, the final 3-D electron density field is reconstructed using the multiplicative algebraic reconstruction technique (MART), as described in Section III-D.

III. Numerical Simulation and Performance Evaluation

To quantitatively evaluate the proposed augmented-state joint tomographic framework, we conduct a series of numerical simulations. The objective is to verify whether the algorithm can decouple heterogeneous hardware biases and suppress spatially varying noise under high-latitude conditions.

A. Semi-Physical Simulation Setup

In order to validate the effectiveness of the proposed multisatellite joint augmented tomographic method, a semi-physical simulation framework is established over the high-latitude Alaska region. Considering that current in-orbit L-band SAR satellites cannot provide time-synchronized and spatially overlapping observations, a virtual collaborative constellation comprising three heterogeneous satellites (Platforms A–C) is constructed based on the actual orbital parameters of PALSAR-1 ("ceos_leader.txt" file).

As shown in Fig. 1, three sets of actual orbital datasets over this region, each comprising 13 along-track scenes, are selected to simulate the different satellite platforms. First, the satellite platform A (Sat A), which represents the PALSAR-2 satellite in this study, utilizes the original descending pass data of PALSAR-1, serving as the baseline observation geometry. Then, the satellite platform B (Sat B), representing the PALSAR-3 satellite, is generated using another set of PALSAR-1 ascending pass data. To simulate the incidence angle diversity inherent to different radar sensors in future heterogeneous constellations, the true latitude and longitude of the illuminated scene center are first extracted from the workreport metadata file associated with each PALSAR scene. By synthetically shifting the ground scene center of the Sat B track eastward by 2° in longitude, a second orbital track is generated to represent the observation geometry of the PALSAR satellite. Finally, the satellite platform C (Sat C), representing the SAOCOM satellite, employs the ascending pass data of PALSAR-1. The combination of ascending and descending passes introduces observation geometries with large crossing angles in 3-D space, significantly increasing the physical probability of ray intersections.

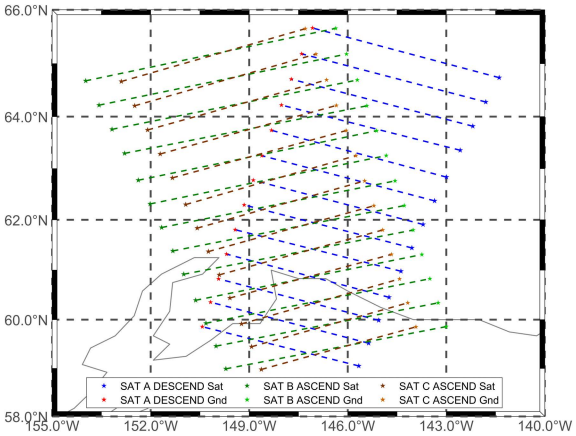


Fig. 1. Plan view of the simulated heterogeneous SAR constellation geometry over a 3° × 2° (latitude/longitude) gridded test site. Dashed lines denote radar lines-of-sight from satellite platforms to ground centers. Blue tracks indicate the descending orbit of PALSAR-2 (Sat A). Green tracks (Sat B) represent synthetic PALSAR-3 observations, whose ground centers are shifted 2° eastward to simulate incidence angle diversity. Brown tracks correspond to the ascending orbit of platform C (Sat C), which simulates the SAOCOM satellite.

Regarding the construction of the true *Ne*, this study avoids over-reliance on idealized smooth models to more authentically reflect the complex morphology at high latitudes (e.g., plasma patch [24], [25]). Instead, the E-CHAIM empirical model is utilized to provide the large-scale background ionosphere (March 31, 2011, 09:00 UTC), onto which localized 3-D electron density enhancement perturbations are synthetically superimposed. All simulated observations are mapped to this same reference ionospheric state. This setting does not imply that existing SAR satellites can provide strictly simultaneous observations; rather, it is used to separate the geometric and calibration effects from the temporal evolution of the ionosphere. As illustrated in Fig. 2, the reconstruction area is discretized into a 3-D voxel grid with a spatial resolution of 3° × 2° × 20 km, in longitude, latitude, and altitude, respectively. This voxel covers a reconstruction volume ranging from 155°W to 140°W in longitude, 58°N to 66°N in latitude, and 100–600 km in altitude. The anomaly is strategically centered at an altitude of 300 km (62.0°N, 144.0°W), and the final 3-D electron density distribution is formulated as the superposition of the background and the anomalous perturbations. This composite environment, featuring abrupt structural variations,

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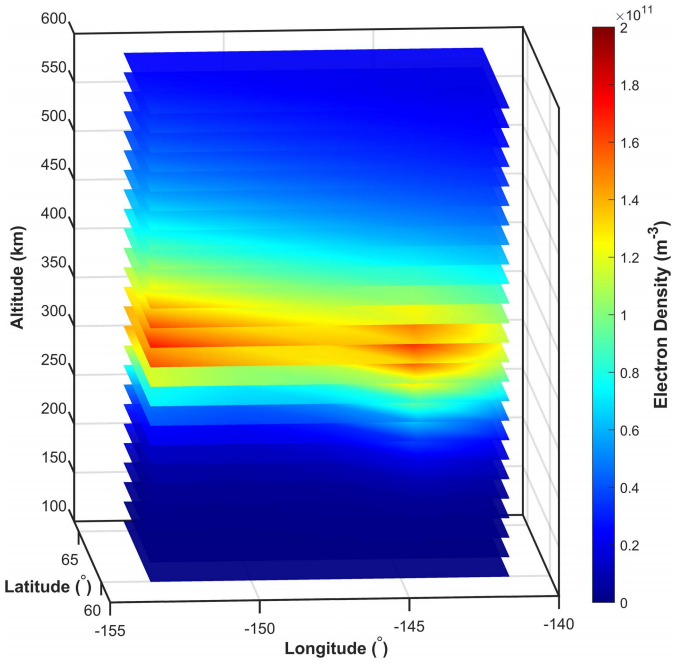


Fig. 2. 3-D true distribution of *Ne*. Slices are rendered from 100 to 600 km with a 20-km vertical step. Background data are derived from the E-CHAIM model (March 31, 2011, 09:00 UTC), with a synthetic enhancement centered at 300 km altitude (62.0°N, 144.0°W) to simulate a localized anomaly.

provides a rigorous benchmark for evaluating the algorithm's resolving power and spatial resolution for fine-scale spatial features.

Then, systematic hardware biases and measurement noise were injected into the three virtual platforms to more authentically simulate the joint observation scenario of a future heterogeneous L-band SAR constellation. Specifically, the three virtual platforms in this study are benchmarked against highly representative L-band SAR systems: PALSAR-2, PALSAR-3, and Argentina's SAOCOM satellite. According to the theoretical derivation of FRA estimation in PolSAR, channel amplitude imbalance typically cancels out during coherent phase extraction. In contrast, δ and ϕ are the core hardware factors that induce systematic biases in FRA and TEC estimations.

Therefore, considering the characteristic differences in antenna arrays and transmit/receive (T/R) modules among various radar sensors, we exclusively introduced platform-specific hardware errors, namely, δ and ϕ, for each heterogeneous platform in the simulation. Furthermore, to emulate realistic electromagnetic signal degradation, independent Gaussian white noise, quantified by the SNR, is injected into all observations. The detailed error injection parameters for each platform are listed in Table I. It should be emphasized that the injected hardware bias parameters in Table I are not arbitrarily assigned, but are comprehensively determined by referring to the nominal system specifications and published in-orbit polarimetric calibration reports of the respective SAR missions. For instance, the parameters for Sat A are benchmarked against the typical polarization isolation baseline (better than −30 dB) of PALSAR-2 [26]; Sat B reflects the stricter error suppression capabilities of the next-generation PALSAR-3 featuring a digital beamforming (DBF) architecture [27]; and Sat C reasonably incorporates the inherent element coupling characteristics typical of SAOCOM's massive phased array antenna evaluated during its commissioning phase [18]. This parameterization, strictly bounded by real-world engineering limits, maximizes the physical credibility of the simulation experiment.

It should be emphasized that the reported performance should be interpreted as the expected behavior under a controlled semi-physical environment, rather than as a direct operational validation using real coordinated multisatellite SAR observations. At present, no public dataset provides strictly synchronized, spatially overlapping, multiangle L-band PolSAR acquisitions over the same high-latitude ionospheric volume, together with an independent 3-D electron density truth for direct benchmarking. Therefore, the semi-physical simulation used in this study is designed as a controlled and reproducible first-step validation. It incorporates real PALSAR-1 orbital parameters and scene geometry, realistic L-band polarimetric error levels, SNR-dependent observation noise, and an E-CHAIM background ionosphere, while keeping the true 3-D electron density known for quantitative evaluation. Future practical validation will require coordinated acquisitions from heterogeneous SAR constellations and independent references such as dense GNSS networks, ionosondes, or incoherent scatter radar measurements.

B. Evaluation of Observation Geometry and Numerical Conditioning

To intuitively illustrate the physical origins of the tomographic ill-posedness and demonstrate the geometric superiority of the proposed multisatellite collaborative framework, the 3-D ray-tracing geometries for both single-satellite and multisatellite configurations are simulated and visualized in Fig. 3.

Fig. 3(a) displays the conventional single-satellite observation geometry (e.g., Sat A). Due to the restricted orbital trajectory and the side-looking imaging mechanism of PolSAR, the penetrating radar rays form a singular, near-parallel observation curtain within the 3-D discrete voxel. Consequently, a substantial number of voxels remain entirely unilluminated, while the traversed voxels are only intersected from a single, highly correlated angular direction.

Mathematically, this lack of angular diversity leads to a severe rank deficiency in the geometric design matrix $\mathbf{G}$, which fundamentally constitutes the ill-posedness of single-pass CIT. In stark contrast, Fig. 3(b) presents the robust ray geometry constructed by the multisatellite collaborative network (Sat A–C). By integrating observations from heterogeneous orbital passes, the distinct ray curtains intersect spatially over the target region. This multiangle overlapping architecture establishes dense geometric constraints within the individual voxels. From an algebraic perspective, these intersecting rays populate the geometric matrix $\mathbf{G}$ with highly linearly independent rows, thereby dramatically reducing the condition number of the unified tomographic equations and effectively mitigating the inherent ill-posedness.

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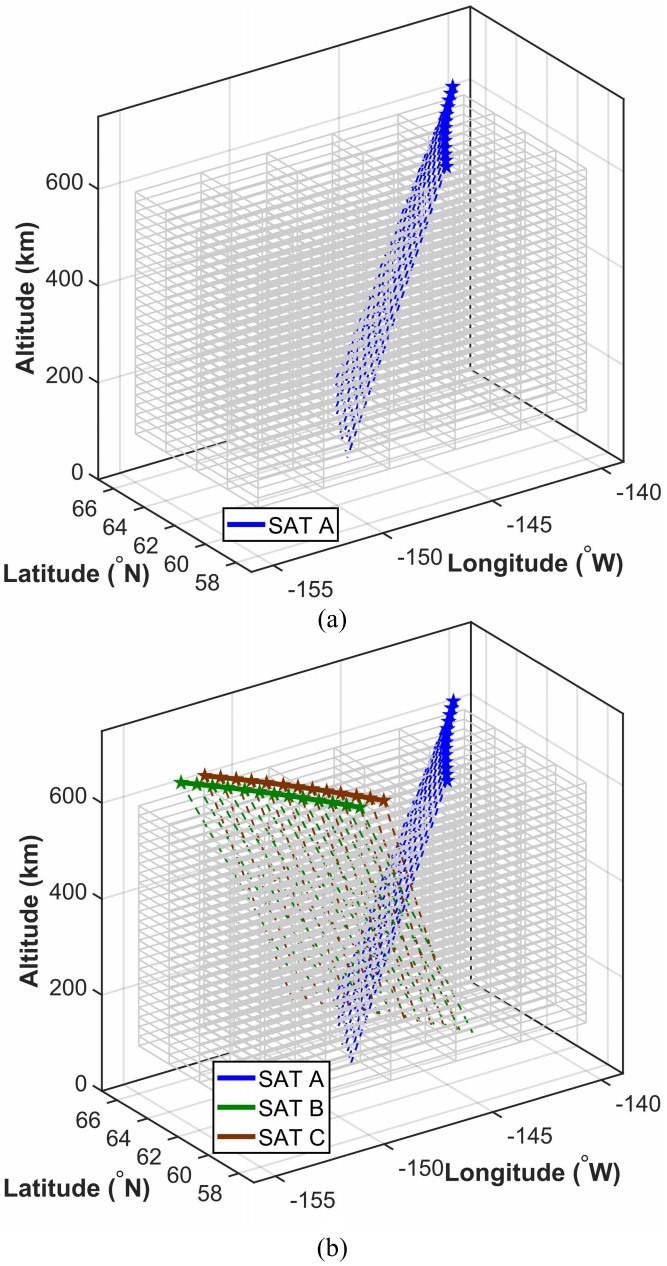


Fig. 3. 3-D visualization of satellite constellation orbital tracks and ionospheric penetration rays in a 2-D display for Fig. 1. (a) 3-D view of the descending satellite Sat A, showing its orbital track and ionospheric penetration rays. (b) Combined 3-D view of three satellite platforms including descending Sat A (blue), ascending Sat B (green), and ascending Sat C (brown).

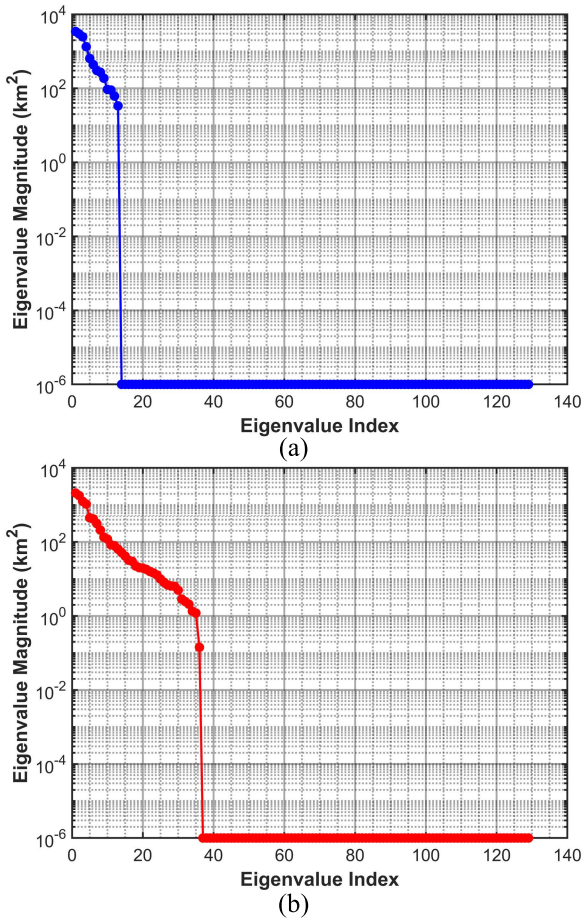


Fig. 4. Eigenvalue spectra of the weakly regularized geometric normal matrices for (a) single-satellite system (Sat A) and (b) multisatellite constellation.

For the numerical comparison, the conditioning of the observation geometry is evaluated using the normalized, weakly regularized geometric normal matrix $A_g = G_s^T G_s / I + \varepsilon I_N$, where G_s contains the columns associated with a fixed set of illuminated voxels, I is the number of observation rays, I_N is the $N \times N$ identity matrix, and $\varepsilon = 10^{-6}$ km^2. This matrix is used only to compare the observation geometries; the weighting matrix $\mathbf{W}$ remains part of the augmented inversion in (21). For the single-satellite configuration, the resulting condition number is 3.42×10^9, indicating a large null-space component, as also reflected by the rapid decay of the eigenvalue spectrum in Fig. 4(a). By incorporating multiangle observations, the condition number is reduced to 2.10×10^9. From a rigorous numerical analysis perspective, absolute magnitudes in the 10^9 regime indicate that the inverse problem remains severely ill-posed and would be dominated by noise under unregularized least-squares estimation. However, in the context of inherently under-determined CIT, this reduction from 3.42×10^9 to 2.10×10^9 corresponds to a factor of 1.63, or approximately a 38.6% decrease in the condition number. Although the absolute condition number remains large, the relative decrease indicates that the multisatellite intersecting geometry reduces the null-space component of the sensing matrix and improves the numerical conditioning of the tomographic system. It serves as a quantitative metric demonstrating that the multisatellite intersecting geometry effectively shrinks the null space of the sensing system [as demonstrated by the extended range of nonzero eigenvalues in Fig. 4(b)]. Numerical stability is further improved by the subsequent Tikhonov regularization and the MART nonnegativity constraint. This geometric improvement also reduces the dependence of the reconstruction on the background prior.

Although a larger voxel size inevitably introduces a smoothing effect that absorbs some microscopic angular differences, this reduction in the condition number indicates that the multisatellite cross-geometry can partially mitigate the severe ill-posedness by reducing the null-space compo-

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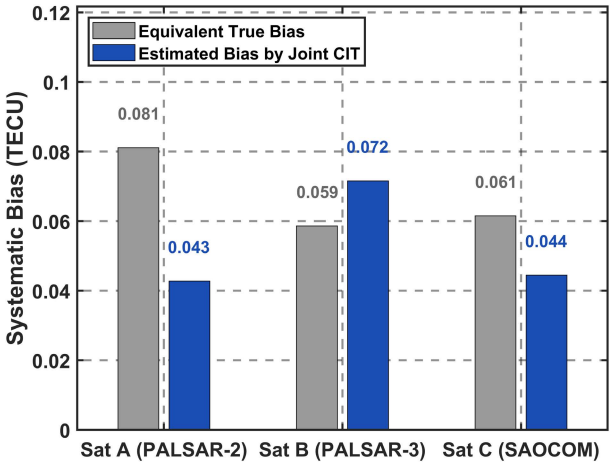


Fig. 5. Quantitative comparison of injected (gray) and estimated (blue) hardware biases. The average residuals (0.023 TECU) illustrate the calibration achieved by the multisatellite joint inversion.

nent of the observation matrix. This improvement reduces the reliance of subsequent iterative algorithms, such as the MART, on the initial background model, suppresses multiplicative artifacts during iteration, and provides a robust numerical foundation for reconstructing 3-D electron density distributions.

C. Heterogeneous Hardware Bias Calibration

In a heterogeneous multisatellite constellation, the SAR payloads on different platforms (i.e., Sat A/PALSAR-2, Sat B/PALSAR-3, and Sat C/SAOCOM) inevitably exhibit varying hardware imperfections. According to the configurations in Table I, these imperfections are primarily characterized by polarimetric crosstalk ranging from −30 to −35 dB and channel phase imbalances of 2°–3°. These hardware distortions, when interacting with the scattering characteristics of natural terrain (assuming $a = 1$ and $\beta = 0.45$ [28], [29]), introduce systematic offsets into the FRA estimation.

To resolve these platform-dependent offsets, the augmented objective function defined in (21) serves as the mathematical core for bias separation. By incorporating the E-CHAIM background model into the regularization term, the framework constrains the inversion to focus on spatially coherent ionospheric perturbations. This strategy effectively mitigates the spectral mixing between the large-scale background TEC and the localized hardware biases, ensuring that the offsets are uniquely identifiable. Furthermore, the weighting matrix $\mathbf{W}$ adaptively accounts for the heterogeneous SNRs (18–22 dB), allowing the estimation process to remain robust against varying observation noise floors.

Specifically, the injected equivalent true biases for Sat A, B, and C are 0.081, 0.059, and 0.061 TECU, respectively. Furthermore, to account for the heterogeneous SNRs defined in Table I (18–22 dB), platform-dependent Gaussian white noise with standard deviations of 0.010, 0.008, and 0.013 TECU are added to the observations of Sat A, B, and C, respectively, representing the residual noise after multilooking processing. The performance of the joint calibration is illustrated in Fig. 5. Specifically, the joint inversion yields the following quantitative outcomes: For Sat A, the injected bias of 0.081 TECU is estimated as 0.043 TECU, resulting in an absolute residual of 0.038 TECU. For Sat B, the estimated value reached 0.072 TECU (true: 0.059 TECU), with a minimal residual of 0.013 TECU. For Sat C, the bias was calibrated to 0.044 TECU (true: 0.061 TECU), with a residual of 0.017 TECU.

To further examine the influence of the relatively larger Sat A residual, a 100-run Monte Carlo test was conducted by varying the random noise realization while keeping the satellite geometry, E-CHAIM background, hardware-error settings, and inversion parameters unchanged. The mean absolute residuals of Sat A, Sat B, and Sat C were 0.013 ± 0.010 TECU, 0.0066 ± 0.0055 TECU, and 0.0071 ± 0.0047 TECU, respectively, with an average residual of 0.0090 ± 0.0051 TECU. The signed residuals averaged over the 100 realizations were close to zero, suggesting that the larger Sat A residual in Fig. 5 mainly reflects a single noise realization rather than a systematic calibration bias. To evaluate its impact on the subsequent 3-D reconstruction, the nominal residual bias vector inferred from Fig. 5 was injected into the MART reconstruction. Compared with the ideal post-calibration case, the global root-mean-square error (RMSE) increased only from 9.816×10^9 to 9.862×10^9 m^{-3}, corresponding to a relative increase of approximately 0.47%. Within the localized anomaly region, the corresponding RMSE increase is approximately 0.20%. This indicates that the remaining calibration residual has a limited influence on the final 3-D reconstruction accuracy.

To further clarify the advantage of the augmented-state formulation over a conventional constrained fusion strategy, a nonaugmented baseline was also considered. In this baseline, the same multisatellite observations and E-CHAIM prior were used, but the platform-dependent bias terms were removed from the state vector. This setting is equivalent to a traditional constrained inversion that assumes all PolSAR-derived TEC observations have already been placed on a consistent absolute scale. In this case, the platform-dependent offsets cannot be estimated and remain as coherent observation biases. The mean magnitude of the injected platform bias is approximately 0.067 TECU. In comparison, the proposed augmented-state inversion reduces the mean residual bias to 0.023 TECU, corresponding to a reduction of 65.7%. This result indicates that the augmented bias terms are important for preventing heterogeneous polarimetric system errors from being absorbed into the reconstructed electron density.

The achievement of an average residual of 0.023 TECU is of paramount importance for the fidelity of 3-D ionospheric imaging. From a signal-processing perspective, the significance of this calibration lies in the optimization of the signal-to-bias ratio (SBR). Prior to calibration, the platform-dependent hardware biases (approximately 0.08 TECU) are of the same order of magnitude as the fine-scale ionospheric anomalies under investigation, which typically exhibit magnitudes between 0.1 and 0.5 TECU. In the context of ill-posed tomographic inversion, such uncompensated systematic errors act as coherent interference, which not only masks authentic geophysical features but also propagates through the sensing matrix to induce significant spatial artifacts.

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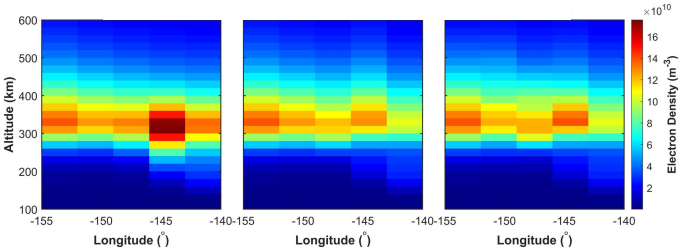


Fig. 6. Vertical cross sections of the electron density distribution at a latitude of 62.0°N. From left to right: the truth profile, the single-satellite reconstruction, and the multisatellite joint reconstruction.

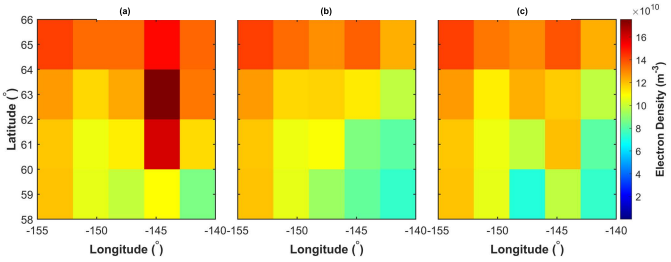


Fig. 7. Horizontal slices of the electron density at the core anomaly altitude of 300 km. From left to right: the truth, the single-satellite reconstruction, and the multisatellite joint reconstruction.

By suppressing the average residual error to 0.023 TECU, the proposed methodology significantly attenuates the structured hardware impairments, reducing them to an order of magnitude comparable to the stochastic thermal noise (0.008–0.013 TECU). While the residual systematic offsets are not entirely eliminated, this substantial reduction ensures that hardware-induced artifacts no longer overwhelmingly dominate the error budget. This level of precision is critical for the unambiguous characterization of sub-TECU ionospheric structures. While the absolute residual remains detectable relative to the true TEC (~3.0 TECU), its suppression is the decisive factor in preventing severe spectral leakage of hardware imbalances into the retrieved electron density distributions. These results indicate that the added geometric constraints and platform-bias terms improve the consistency of the multiplatform observations used for CIT.

D. 3-D Electron Density Reconstruction Performance

Building upon the successful calibration of platform-dependent hardware biases achieved in Section III-C, this section evaluates the ionospheric reconstruction performance of the multisatellite constellation. With the systematic errors accurately suppressed, the MART is employed to iteratively reconstruct the 3-D electron density [30], [31]. MART is particularly well-suited for this highly under-determined tomographic problem, as it naturally enforces the physical non-negativity constraint of the electron density while effectively handling the sparse geometric sensing matrix.

To evaluate reconstruction fidelity, a synthetic localized enhancement centered at 300 km (62.0°N, 144.0°W) is superimposed onto the E-CHAIM background model to serve as the truth (Fig. 2). In the ensuing analysis, the reconstructed 3-D electron density field is visualized through horizontal slices along the altitude profile. This slicing strategy directly mirrors the truth visualization, facilitating an intuitive and quantitative comparison of the morphological recovery at varying altitudes.

To demonstrate the superiority of the multisatellite constellation, a comparative experiment using single-satellite observations (e.g., Sat A only) is conducted. As illustrated in the vertical cross sections (Fig. 6), the single-satellite reconstruction suffers from vertical smearing. Due to the limited viewing angles and parallel ray paths, the localized electron density anomaly is artificially elongated along the elevation direction, failing to resolve the true altitude gradient.

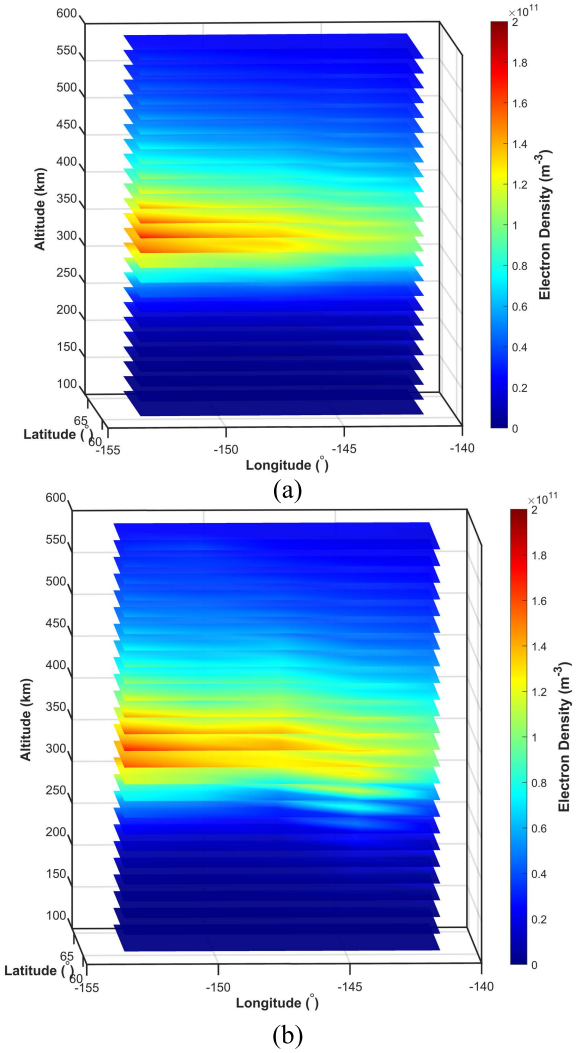


Fig. 8. CIT result based on (a) single Sat A and (b) multisatellite constellation (Sat A–C).

By incorporating the intersecting geometry of the multisatellite network, the joint inversion effectively mitigates this smearing effect. This enhancement is visually corroborated by the horizontal cross section at 300 km (Fig. 7), where the multisatellite output delineates the spatial morphology of the anomaly closer to the truth than the single-satellite estimation. Finally, the comprehensive 3-D stacked slices further highlight this contrast; while the single-satellite result [Fig. 8(a)] exhibits

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severe artificial elongation, the multisatellite configuration [Fig. 8(b)] concentrates the electron density closer to the genuine 300 km altitude, demonstrating a notable overall improvement in spatial resolution.

To further quantify the improvement in spatial localization and vertical-smearing suppression, several additional metrics were calculated from the reconstructed anomaly relative to the E-CHAIM background. Compared with the single-satellite result, the multisatellite reconstruction increases the correlation coefficient from 0.9706 to 0.9769 and reduces the global RMSE from 11.16×10^9 to 9.79×10^9 m^{-3}. The peak recovery ratio of the localized anomaly increases from 0.300 to 0.781, indicating that the multisatellite geometry better preserves the anomaly amplitude. At the true anomaly column, the vertical centroid error decreases from 110.4 to 89.9 km, and the vertical full width at half maximum (FWHM) decreases from 120 to 100 km. In addition, the vertical leakage ratio outside the main anomaly altitude range decreases from 0.399 to 0.354. These quantitative metrics support the visual comparison in Figs. 6 and 7, showing that the proposed multisatellite configuration reduces vertical smearing and improves the localization of the reconstructed ionospheric perturbation, although residual smoothing still remains.

Quantitatively, the geometric advantage of the multisatellite constellation is clearly reflected in the global error metrics evaluated over all CIT voxels against the true electron density distribution (Fig. 2). While both configurations benefit from the E-CHAIM prior to maintain general background consistency, the single-satellite inversion yields a global RMSE RMSE of 1.12×10^{10} m^{-3} due to the substantial numerical errors induced by the aforementioned vertical smearing artifacts [Fig. 8(a)]. The multisatellite framework effectively suppresses these geometric distortions, reducing the global RMSE to 9.79×10^9 m^{-3} [Fig. 8(b)]. This represents a 12.2% overall reduction in reconstruction error. The discrepancy between the true electron density in Fig. 2 and the reconstructed results in Fig. 8 can be explained by the severely under-determined nature of spaceborne SAR CIT. The synthetic truth contains a localized high-gradient ionospheric enhancement, whereas the available SAR observations provide only a limited number of slant TEC integrations through the 3-D voxel grid. Therefore, a considerable portion of the electron density field remains weakly constrained by the observation geometry. Under this condition, the reconstruction is not expected to exactly recover the pointwise peak density. The E-CHAIM prior, Tikhonov stabilization, and MART nonnegative constraint jointly improve numerical stability, but they also tend to distribute the correction over neighboring illuminated voxels, leading to an attenuated peak amplitude and a smoother anomaly core. Hence, the remaining difference between Figs. 2 and 8 mainly reflects the limited angular diversity, finite voxel resolution, and regularization-induced smoothing of the current observation geometry. The proposed multisatellite framework reduces the vertical smearing and improves the global reconstruction accuracy, but it does not completely remove the null-space components of the ill-posed tomographic problem.

TABLE II
SENSITIVITY OF THE MULTISATELLITE RECONSTRUCTION TO THE TIKHONOV REGULARIZATION COEFFICIENT

$\lambda/\lambda 0$	λ	Mean Bias Residual (TECU)	Multi-Satellite RMSE ($10^{10} m^{-3}$)	Main Observation
0.001	1.0×10^{-5}	0.0600	0.986	Weak prior; larger bias residual
0.01	1.0×10^{-4}	0.0347	0.984	Weak prior; larger bias residual
0.1	1.0×10^{-3}	0.0243	0.983	Near stable range
1	1.0×10^{-2}	0.0236	0.983	Selected $\lambda 0$
10	1.0×10^{-1}	0.0235	0.983	Stable RMSE
100	1.0	0.0233	0.983	Stable RMSE
1000	10	0.0217	0.982	Stable RMSE

To examine whether this improvement depends on a particular choice of λ, a sensitivity test was further conducted. In this test, the satellite geometry, E-CHAIM background, injected hardware biases, noise level, and reconstruction settings were kept unchanged, while λ was varied from $0.001\lambda 0$ to $1000\lambda 0$. The results are listed in Table II. When λ is much smaller than $\lambda 0$, the mean bias residual increases, indicating that the prior constraint is too weak to sufficiently separate the ionospheric perturbation from the platform-dependent bias. Around and above $0.1\lambda 0$, the multisatellite RMSE remains nearly unchanged, ranging from $0.982 \sim 0.983 \times 10^{10}$ m^{-3}. This indicates that the reconstruction improvement is mainly associated with the enhanced multisatellite observation geometry rather than an arbitrary tuning of the regularization coefficient. The small RMSE variation over this range indicates that the reported reconstruction improvement is not sensitive to the selected value of λ.

Although the proposed framework substantially mitigates the vertical smearing, residual over-smoothing is still observed at the anomaly core in the multisatellite result [Fig. 8(b)], where the absolute peak intensity is not fully recovered to the exact truth magnitude. This is fundamentally constrained by the specific orbital geometry of the current constellation. While Sat A provides a crucial intersecting view, the ascending orbits of Sat B and C are nearly parallel, resulting in highly redundant geometric constraints and limiting the ultimate angular diversity. Future deployments of heterogeneous SAR mega-constellations, featuring greater variations in orbital inclinations, heading angles, and incidence angles, may provide more diverse intersecting ray paths. Such enhanced angular diversity is theoretically expected to further reduce geometric artifacts and improve the recovery of peak intensity.

To further quantify this expectation, a controlled track-rotation experiment is conducted. The current A+B+C constellation was used as the reference geometry. The two controlled cases were generated by rotating the companion track families by $\pm 30^\circ$ and $\pm 60^\circ$, respectively, relative to the Sat A reference track in the local horizontal plane. These idealized cases were designed to isolate the effect of angular diversity rather than

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TABLE III
CONTROLLED TRACK-ROTATION EXPERIMENT FOR MULTISATELLITE GEOMETRIES

Scenario	Nominal Track Rotation	Valid Voxels	Condition Number ($\times 10^9$)	Effective Rank	RMSE (10^10 m^-3)	RMSE Reduction
Current A+B+C	Current	218	2.10	26.19	0.979	0
Controlled ±30°	±30°	271	2.40	28.93	0.903	7.8%
Controlled ±60°	±60°	303	1.88	29.61	0.815	16.7%

to represent a complete orbital-design study. In this test, the voxel grid, E-CHAIM background, synthetic anomaly, noise level, number of rays, and MART reconstruction parameters were kept unchanged, so that the comparison mainly reflects the influence of observation geometry. The effective rank is defined as $r_{\text{eff}} = \exp\left(-\sum_j p_j \ln p_j\right)$, where

$p_j = \sigma_j / \sum_k \sigma_k$, and σ_j denotes the singular values of the illuminated geometric matrix. A valid voxel is defined as a voxel intersected by at least one observation ray. The results are summarized in Table III. Compared with the current geometry, the ±30° and ±60° track-rotation cases increase the number of valid voxels from 218 to 271 and 303, respectively, and improve the effective rank from 26.19 to 28.93 and 29.61. The corresponding RMSE is reduced from 9.79×10^9 to 9.03×10^9 m^{-3} and 8.15×10^9 m^{-3}, respectively. The condition number does not vary monotonically with the nominal track rotation because it is also affected by the discrete voxel-intersection pattern. Nevertheless, under this controlled setting, the improvement in effective rank, spatial coverage, and RMSE suggests that larger angular diversity can provide less redundant geometric constraints and may further mitigate the residual vertical smearing observed in the present constellation.

GNSS-SAR fusion tomography can provide useful external constraints when ground-based GNSS or ionosonde observations are available [6], [14]. The present work addresses a different scenario, namely high-latitude PolSAR CIT where such external references are sparse or unavailable. Therefore, the proposed augmented-state strategy is not intended to replace GNSS-SAR fusion, but to provide a self-calibrated PolSAR-only framework for future multisatellite SAR constellations. In principle, the same augmented formulation can also be extended to GNSS-SAR fusion by adding GNSS ray-path observations to the geometric matrix and assigning appropriate bias or weighting terms to different data sources.

IV. CONCLUSION

This study presented a novel joint 3-D reconstruction framework for low-frequency fully PolSAR ionospheric tomography based on a multisatellite constellation. To overcome the severe ill-posedness and vertical smearing artifacts inherent in conventional single-satellite CIT, we established a semi-physical simulation framework and proposed an augmented state vector approach to adaptively calibrate heterogeneous platform-dependent polarimetric system biases. Quantitative evaluations demonstrated that the multisatellite intersecting observation geometry measurably improves the geometric observability of the tomographic equations, decreasing the dimensionless condition number from 3.42×10^9 to 2.10×10^9, corresponding to a factor of 1.63 or a 38.6% decrease. Consequently, the proposed joint inversion framework effectively suppressed the geometric artifacts, yielding a 12.2% reduction in the RMSE compared to the single-satellite model.

Although the proposed framework substantially mitigates severe vertical smearing, the residual over-smoothing observed at the anomaly core indicates that the current orbital geometry still imposes fundamental limitations. In particular, the nearly parallel ascending orbits of Sat B and C provide partially redundant geometric constraints, limiting the available angular diversity. The additional controlled track-rotation experiment further suggests that future constellations with larger angular diversity can improve voxel coverage and effective rank, leading to RMSE reductions of 7.8% and 16.7% for the ±30° and ±60° track-rotation cases, respectively, under the controlled simulation setting. This result supports the expectation that heterogeneous SAR constellations with more diverse orbit inclinations, heading angles, and incidence angles may further suppress residual geometric artifacts and improve the recovery of localized ionospheric perturbations. Overall, this study demonstrates the potential of distributed SAR networks for space weather monitoring and provides algorithmic and theoretical references for 3-D ionospheric reconstruction in future L/P-band multisatellite constellations. Nevertheless, practical validation using real coordinated multisatellite PolSAR observations and independent ionospheric reference measurements remains an important next step once suitable synchronized datasets become available.

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APPENDIX

A. Polarimetric Crosstalk

Let δ represent the symmetric complex crosstalk coefficient (with $\delta \ll 1$). At the physical hardware level, the true scattering signal of the target is contaminated by the crosstalk matrix $\mathbf{C} = \begin{bmatrix} 1 & \delta \\ \delta & 1 \end{bmatrix}$. By expanding the actual measurement matrix $\mathbf{M}_\delta = \mathbf{C^T M C}$ and omitting the negligibly small second-order system parameter terms (i.e., $\delta^2 \approx 0$), it can be observed that the weak cross-polarization channel is dominated by the energy leakage from the strong co-polarization signals (S_{HH} and S_{VV}). Projecting this into the circular polarization basis, the perturbation increment ΔZ_{LR} of the contaminated circular cross-polarization measurement is approximated as

$$\Delta Z_{LR} = \Delta Z_{RL} \approx -\text{j}\delta\,(S_{HH} - S_{VV})\,. \tag{A1}$$

Defining the co-polarization ratio α of the target as

$$\alpha = \frac{S_{HH} - S_{VV}}{S_{HH} + S_{VV}}. \tag{A2}$$

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By definition, it follows that $S_{HH} - S_{VV} = \alpha(S_{HH} + S_{VV})$. Since the sum of the co-polarization energy has been explicitly defined as equivalent to $S_{HH} + S_{VV} = -2\mathrm{j}Z_x$ in (3), the perturbation term can be equivalently expressed as $\Delta Z_{LR} \approx -2\,\delta a Z_x$. Consequently, the contaminated B&B core product becomes $\hat{A} = (Z_{LR} + \Delta Z_{LR})(Z_{RL} + \Delta Z_{RL})^*$. Expanding this polynomial and discarding the negligibly small second-order cross terms based on the first-order complex perturbation theory (i.e., the product of infinitesimals $\Delta Z_{LR}\Delta Z_{RL}^* \approx 0$), we obtain $\hat{A} = A + \varepsilon$, where the core first-order perturbation term is rigorously expressed as

$$\varepsilon = Z_{LR}\Delta Z_{RL}^* + \Delta Z_{LR} Z_{RL}^*. \tag{A3}$$

Dividing this by the ideal, unpolluted product $A = Z_{LR}Z_{RL}^*$, substituting the exponential forms of the ideal circular polarization terms (3), and utilizing the complex perturbation expansion along with conjugate symmetry, we derive the complex perturbation ratio of the B&B core product A as

$$\frac{\varepsilon}{A} \approx -4Re(\delta\alpha)\, e^{-j2\Omega}. \tag{A4}$$

Applying the small-angle phase approximation theorem (which states that for an infinitesimally small complex perturbation x, $\arg(1 + x) \approx Im(x)$, we have

$$\Delta \arg\left(\hat{\mathbf{A}}\right) \approx Im\left(\frac{\varepsilon}{A}\right). \tag{A5}$$

Substituting (A4) into (A5), applying Euler's formula $e^{-\mathrm{j}2\Omega} = \cos(2\,\Omega) - \mathrm{j}\sin(2\,\Omega)$, and extracting the imaginary part yields $\Delta\arg(\hat{\mathbf{A}}) \approx 4Re(\delta\alpha)\sin(2\,\Omega)$. Considering that extracting the FRA requires a division by 4 [see (4)], the final systematic bias $\Delta\Omega_{cross}$ induced by crosstalk is strictly equivalent to

$$\Delta\Omega_{cross} \approx Re(\delta\alpha)\sin(2\,\Omega). \tag{A6}$$

B. Channel Imbalance

Let Δ_A denote the relative imbalance of the V channel with respect to the H channel. In the complex perturbation analysis framework, the equivalent complex perturbation of the V channel in the linear basis is modeled with an additional 90° phase projection (i.e., j) to strictly separate amplitude and phase imbalances in subsequent algebraic operations

$$\Delta M_{VV} = \mathrm{j}\Delta_A S_{VV}. \tag{A7}$$

When projecting the linear matrix $\mathbf{M}$ into the circular basis, the perturbation in the cross-polarization term ΔZ_{LR} is solely contributed by ΔM_{VV} with a mapping coefficient of j/2

$$\Delta Z_{LR} = \frac{\mathrm{j}}{2}\Delta M_{VV} = \frac{\mathrm{j}}{2}\left(\mathrm{j}\Delta_A S_{VV}\right) = -\frac{1}{2}\Delta_A S_{VV}. \tag{A8}$$

Since $\mathrm{j}^2 = -1$ in complex arithmetic, the induced perturbation in the circular basis is strictly derived as

$$\Delta Z_{LR} = \Delta Z_{RL} = -\frac{1}{2}\Delta_A S_{VV}. \tag{A9}$$

By setting the target parameter $\beta = S_{VV}/(S_{HH} + S_{VV})$, we obtain $S_{VV} = -2\mathrm{j}\beta Z_x$. The perturbation becomes

$$\Delta Z_{LR} \approx -\frac{1}{2}\Delta_A\left(-2\mathrm{j}\beta Z_x\right) = \mathrm{j}\Delta_A\beta Z_x. \tag{A10}$$

Let $Q = \mathrm{j}\Delta_A\beta$. Adopting the identical complex product expansion strategy used for crosstalk, the complex perturbation ratio becomes $2Re(Q)e^{-\mathrm{j}2\Omega}$.

Amplitude Imbalance: For natural distributed targets, the co-polarization phase difference is generally negligible, rendering the target parameter β a real number. If the system exhibits a pure polarimetric amplitude imbalance (i.e., $\Delta_A = A_m$, where A_m is a real number), the variable Q becomes a purely imaginary number. Consequently, its real part vanishes ($Re(Q) = 0$). The induced systematic bias is mathematically strictly evaluated as

$$\Delta\Omega_{amp} = -\frac{1}{2}\mathrm{Re}\left(\mathrm{j}A_m\beta\right)\sin(2\,\Omega) = 0. \tag{A11a}$$

This explicitly provides a rigorous algebraic proof of the B&B algorithm's absolute theoretical immunity to pure amplitude imbalance.

Phase Imbalance Sensitivity: Conversely, if the system exhibits a residual phase imbalance ($\Delta_A = \mathrm{j}\phi$, where ϕ is the real phase error), Q becomes a real number ($Q = \mathrm{j}(\mathrm{j}\phi)\beta = -\phi\beta$), resulting in $\mathrm{Re}(Q) = -\phi\beta$. The phase error becomes

$$\Delta(4\,\Omega) = -2(-\phi\beta)\sin(2\,\Omega) = 2\phi\beta\sin(2\,\Omega). \tag{A11b}$$

Consequently, the systematic bias introduced by phase imbalance is

$$\Delta\Omega_{imb} \approx \frac{1}{2}\phi\beta\sin(2\,\Omega). \tag{A12}$$

Synthesizing both effects, the compound systematic bias for the ith radar ray observed by a specific heterogeneous platform is rigorously defined as

$$B_{\Omega,i} \approx \left[Re(\delta_i\alpha) + \frac{1}{2}\phi_i\beta\right]\sin(2\,\Omega) \tag{A13}$$

which correctly corresponds to (12) in the main text.

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